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API and JSONs

Assignment

Understanding APIs for Corporate Employees

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TOTAL QUESTIONS 30	DURATION 60 Minutes	TOTAL MARKS 70

Instructions

- This assignment contains 30 questions across five sections: Multiple Choice, True/False, Fill in the Blanks, Short Answer, and Practical/Coding.
- Type your answers directly into this document — in the blank fields, answer lines, or code boxes provided under each question.
- You may refer to official documentation, but all answers must be written in your own words / own code.
- Total marks: 70. Suggested duration: 60 minutes.
- Where an API endpoint is needed, use <https://jsonplaceholder.typicode.com> unless another URL is specified.

Section A — Multiple Choice Questions

10 marks

Choose the single best answer. (1 mark each)

1. What does API stand for?

- A) Application Programming Interface
- B) Automated Program Integration
- C) Application Process Interaction
- D) Advanced Programming Instruction

Your Answer: ____A____

2. In the client–server model, which of these is typically the “client”?

- A) A database
- B) A web browser or mobile app
- C) A web server
- D) An operating system kernel

Your Answer: ____B____

3. Which HTTP method is used to retrieve data without modifying it?

- A) POST
- B) GET
- C) DELETE
- D) PUT

Your Answer: ____B____

4. What does an HTTP status code of 404 mean?

- A) Internal server error
- B) Unauthorized access
- C) Resource not found
- D) Request succeeded

Your Answer: ____C____

5. Which HTTP method replaces an entire existing resource?

- A) PATCH
- B) PUT
- C) GET
- D) POST

Your Answer: ____B____

6. What data format do most modern web APIs use to exchange data?

- A) XML only
- B) CSV
- C) JSON
- D) Plain text

Your Answer: _____C_____

7. Which status code indicates a resource was successfully created?

- A) 200
- B) 201
- C) 301
- D) 404

Your Answer: _____B_____

8. In Python, which library is most commonly used to send HTTP requests?

- A) pandas
- B) requests
- C) numpy
- D) flask

Your Answer: _____B_____

9. What does `response.json()` do?

- A) Sends JSON data to the server
- B) Converts the response body into a Python dictionary
- C) Permanently deletes JSON data
- D) Encrypts the JSON payload

Your Answer: _____B_____

10. Which HTTP header is typically used to send authentication credentials?

- A) Content-Type
- B) Authorization
- C) Accept
- D) User-Agent

Your Answer: _____B_____

Section B — True or False

5 marks

Type True or False. (1 mark each)

11. A GET request can safely be sent multiple times without changing data on the server.

Your Answer (True / False): _____True_____

12. JSON stands for JavaScript Object Notation.

Your Answer (True / False): _____True_____

13. A Python dictionary and a JSON object are exactly the same thing in every way.

Your Answer (True / False): _____False_____

14. PATCH updates an entire resource, exactly like PUT.

Your Answer (True / False): _____False_____

15. A `requests.Session()` allows multiple requests to reuse the same underlying connection.

Your Answer (True / False): _____ True _____

Section C — Fill in the Blanks

5 marks

Type the missing word or term. (1 mark each)

16. The HTTP status code _____ means “Unauthorized.”

Your Answer: _____ 401 _____

17. The _____ HTTP method is used to permanently remove a resource from a server.

Your Answer: _____ DELETE _____

18. In a URL, the part after the domain that identifies a specific resource is called the _____.

Your Answer: _____ PATH _____

19. The Python function requests. _____ () is used to send a GET request.

Your Answer: _____ GET _____

20. A _____ parameter is used to filter or customize a GET request without changing the endpoint itself.

Your Answer: _____ query _____

Section D — Short Answer Questions

15 marks

Answer in 2–4 sentences. (3 marks each)

21. Explain the difference between PUT and PATCH, and give one example of when you would use each.

- PUT updates the entire record.
- PATCH updates only specific fields.

Example : Use put to replace a whole user profile, and patch to change only the email address or any one field.

22. Describe the request–response cycle in your own words, using a real-life analogy (for example, a restaurant or a pizza order).

The request–response cycle is like ordering a pizza. You place an order with the restaurant (request), the restaurant prepares the pizza, and then delivers it back to you (response).

In the same way, a client sends a request to a server, and the server processes it sends back the required data. This is a request-response cycle.

23. Why is it unsafe to hardcode an API key directly in your code? What is the safer alternative?

Hardcoding an API key is unsafe because anyone who sees the code can steal and misuse the key.

A safer alternative is to store the API key in environment variables or a secure location or in a .env file to protect data.

24. What is the purpose of setting a timeout on an API call? What happens if you don't set one?

Timeout is the maximum time a program waits for a response from a server. If the server takes too long, the request is stopped and an error is raised.

Without a timeout, the program may keep waiting and may also hang if the server is unavailable.

25. Explain what a Session object does in the requests library and why it can improve performance.

A Session object allows multiple requests to reuse the same connection again .

This helps to improve performance because python does not need to create a new connection for every request everytime, making the api calls faster and efficient.

Section E — Practical / Coding Questions

20 marks

Type working Python code into the box below each question.

26. Write Python code using the requests library to send a GET request to <https://jsonplaceholder.typicode.com/posts> and print the response status code.

```
import requests
url = "https://jsonplaceholder.typicode.com/posts"
result = requests.get(url)
print("Response Status:", result.status_code)
```

27. Write Python code to send a POST request to <https://jsonplaceholder.typicode.com/posts> with a JSON body containing a “title” and a “body” field, then print the JSON response.

Hint: Use `json=your_dictionary` as the second argument to `requests.post()`.

```
import requests
url = "https://jsonplaceholder.typicode.com/posts"
post_data = {
    "title": "Neeharika",
    "body": "This is a sample post request."
}
response = requests.post(url,json=post_data)
print(response.json())
```

28. . Write Python code that uses query parameters to fetch all posts where `userId` equals 2.

Hint: Build a `params` dictionary and pass it with `params=params`.

```
import requests
url= "https://jsonplaceholder.typicode.com/posts"
params = {
    "userId": 2
}
response = requests.get(url,params=params)
print("Posts of user 2:")
print(response.json())
```

29. Write a `try/except` block that sends a GET request with a 5-second timeout, and prints “Request Timed Out” if the server does not respond in time.

Hint: Catch `requests.exceptions.Timeout`.

```
import requests
try:
    url = "https://jsonplaceholder.typicode.com/posts"
    result = requests.get(url, timeout=9)
    print("Request is successful ! ")
    print(result.status_code)
except requests.exceptions.Timeout:
    print("Request has timed out ... ")
```

30. Convert the JSON response from <https://jsonplaceholder.typicode.com/posts> into a pandas DataFrame and display the first 5 rows.

Hint: Use `pd.DataFrame(response.json())` and `.head()`.

```
import requests
import pandas as pd
response = requests.get("https://jsonplaceholder.typicode.com/posts")
data = response.json()
records = pd.DataFrame(data)
print("First five rows:")
print(records.head())
```

— End of Assignment —